

Revised ePC Experimental Programme: Homotopy Absorption, Fibre Emergence, and the Deployment Settle

Working plan v1.1 — mid-flight revision, for implementation by coding agents

Causal Fibres / Hyperon ePC effort

July 23, 2026 (supersedes v1.0 of the same date)

Abstract

Version 1.1 revises the v1.0 programme in light of clarifications and new measurements from Mesto (author correspondence, July 23, 2026) that resolve both ambiguities v1.0 was designed to guard against, and that materially change the interpretation of the homotopy result. The central update: at 124M the settled ePC error field is, to high precision, the backpropagation adjoint field — all fidelity numbers were teacher-free feedforward all along, the errors are absolutely small (5.6×10^{-4} global L2), and the “error-norm ratio” curve is a gradient-alignment probe ($\|\text{PC grad}\|/\|\text{BP grad}\|$ rising $\eta \rightarrow 0.98$ at per-block cosine ≥ 0.998). Consequences for this programme: the emergence battery (E2) gains a cheap adjoint-field track that requires no homotopy and starts immediately; E3 gains gradient-sparsification baselines and a proximal-step ablation; E4 is reframed — there is no amortization gap because nothing is amortized, and the entire deployment question reduces to whether test-time information injection beats the feedforward default; E1’s interesting content shifts from “does the homotopy avoid collapse” (Mesto’s depth profile says it should) to localizing what in the R8/R9 recipe caused collapse. Section 1 tells implementing agents exactly what to do with completed (E1) and in-flight (E2) work; the remainder restates the full programme so this document is self-contained and v1.0 can be retired.

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1 Revision memo for implementing agents (v1.0 $\rightarrow$ v1.1)

Status assumed: E1 is complete under v1.0; E2 is in flight under v1.0. Nothing already run is wasted; this section lists the deltas, in priority order. Where v1.1 conflicts with v1.0, v1.1 governs.

1.1 Why the revision

Mesto's reply corrects two misreadings of his report and adds new measurements:

1. **All fidelity numbers were teacher-free.** Evaluation in his report is plain feedforward ($\varepsilon=0$) on both student and teacher; the teacher enters only the training-time settle objective. The v1.0 branch E1-C scenario (clamped-only fidelity) is off the table for his result.

2. **The rising curve is a gradient probe, not a state-space quantity.** The “error-norm ratio” is $\|\nabla_W E_W\|/\|\nabla_W^{\text{BP}} \mathcal{L}\|$ per block, teacher-free under cross-entropy: it starts at exactly η (the $T=1$ anchor identity, observed to the digit) and rises to 0.98, with per-block cosine between the two gradients never below 0.998. Together with the tiny error norms (5.6×10^{-4} global L2 over 4.72M cells), the settled error field $\approx$ the BP adjoint field: *the homotopy interpolates the magnitude of the update from an η -scaled BP step to a full-magnitude local update that never leaves BP’s direction.*
3. **Depth profile (new data).** Top-5% frontier mass by block (1–12, % of frontier cells): 11.1, 18.7, 17.2, 14.8, 15.3, 11.3, 7.2, 2.8, 1.5, 0.1, 0.1, 0.0. Blocks 2–5 hold 66%; the last three blocks hold 0.2%. This is the *opposite* of the R8/R9 late-block pile-up.
4. **Objective-swap robustness (new data).** Replacing the KD term with plain cross-entropy on data (no teacher anywhere) preserves the structure: top-5% share 91.6% vs. 90.0%, same depth shape (13.1, 20.8, 19.5, 15.9, 12.6, 8.2, 6.2, 2.4, 0.9, 0.2, 0.1, 0.0). Across fresh samples under both objectives the top-5% share stays in 82–97%; the fixed-depth Hill statistic moves 1.38–2.00 and is a tail-weight descriptor only.
5. **Deployment is trivially resolved in his frame.** With the task term absent, the settle energy is the pure quadratic in ε , minimized at $\varepsilon=0$: the deployed artifact is exactly the feedforward student. PC in this construction is a learning-time device; nothing is amortized.

1.2 Interpretive frame carried through the rest of the document

At the terminal homotopy, *PC* is a local iterative scheme that computes backpropagation’s credit signal. This is a clean confirmation of the PC-approximates-BP correspondence at $10^3 \times$ prior scale, and it has two programme-level corollaries. (a) The concentration measurement is a measurement of **adjoint sparsity**, obtainable on any trained model with plain autograd, no homotopy required — which is why E2 gains an adjoint-field track that starts immediately. (b) If the credit signal is the same signal, the PC parameterisation *alone* should confer approximately no continual-learning advantage; any advantage must enter via frontier gating (the S3/S4 lesson) or via the one residual structural difference, the proximal/implicitly-regularized character of a settle-based update versus a raw gradient step. Both corollaries are pre-registered as Predictions 1 and 2 below.

1.3 Action list, in priority order

1. **[E2, in flight — do not discard]** Keep all v1.0 E2 work on the settled error field. *Add* the adjoint-field track (§5): compute plain BP adjoint fields (teacher-free CE) on the *same* persisted probe set and run the identical M1–M5 battery on them. Add the new equivalence diagnostic M0 (error field vs. adjoint field, per block) — on our rig this verifies whether Mesto’s 124M identification holds at six layers, and it determines how the two tracks’ results are to be read against each other.
2. **[E2]** Pre-register the co-location prediction (Prediction 2) before unblinding fingerprint results: if factor-aligned structure exists, it should concentrate in the frontier’s home blocks (early-to-middle by Mesto’s profile; use our own measured profile).
3. **[E1, complete]** Record the branch taken under v1.0 criteria in the phase memo, then run the two cheap post-hoc checks in §4: (i) our own frontier depth profile vs. Mesto’s, KD and CE variants; (ii) M0 on the E1 terminal student. Queue the new optional arm E1b (homotopy

schedule *with* the energy-monotonicity term restored) at low priority — it localizes the cause of the R8/R9 collapse but blocks nothing.

4. **[E3]** Add two conditions: a gradient-sparsification baseline (top- k with error feedback / momentum correction, DGC-style) as the externally-tuned null model for frontier settles; and a proximal ablation (settle-based update vs. raw BP step at matched effective step size) isolating the implicit-regularization difference. Details in §6.
5. **[E4]** Reframe per §7: FF is the deployed default by construction; drop “amortization gap” framing from configs, logs, and memos; the experiment now purely measures test-time information injection (SC, OT) against FF, with TC retained as a diagnostic upper anchor only.
6. **[E5]** No structural change; W3’s frontier writes are now to be documented and benchmarked as sparsified-gradient writes (§8).
7. **[Harness]** New shared components: adjoint-field extractor writing to the same artifact schema as settled error fields; M0 evaluator; update the acceptance-criteria JSON to the Appendix A v1.1 table (version bump, keep v1.0 file in place for the E1 record).

1.4 Pre-registered predictions (freeze before the relevant unblinding)

Prediction 1. In E3-V4 and any continual-learning comparison, the PC parameterisation alone (dense settle, no gating) shows no advantage over matched-compute, matched-seed BP fine-tuning beyond noise; advantages, if any, appear only in gated/support-restricted conditions or trace to the proximal ablation.

Prediction 2. If E2 finds factor-aligned structure (M3/M4 above null), the aligned blocks co-locate with the frontier’s home blocks (the depth range holding the majority of top-5% mass), not with the readout-adjacent blocks.

2 Context (revised)

2.1 Prior internal results assumed known

1. **R8/R9 (absorption collapse).** Direct ePC distillation of the six-layer teacher under energy-monotonicity training concentrated update credit in the final block and collapsed representation rank; this motivated the frozen-backbone + sidecar architecture.
2. **S2.** Energy descent can coexist with causally wrong credit; correctness is established inter-ventionally, never from energy curves.
3. **S3/S4.** With supplied supports/fibres in canonical coordinates, fibre-local plasticity gives zero unrelated drift; transports are identified separately from routing. All positive evidence is supplied-structure.
4. **v0.3 JBD.** Robust joint block diagonalization with repeated-copy underidentification flagging.

2.2 External input: Mesto (2026), as now understood

1. **Homotopy distillation, 124M.** GPT-2 deformed into ePC along $\tau = \eta T$, $T \in \{1, \dots, 128\}$, BP-anchored at $T=1$. Teacher-free feedforward fidelity: worst milestone KL 1.3×10^{-5} nats over five runs including terminal τ ; held-out perplexity within 0.35%. Settled error field 5.6×10^{-4} global L2: *the weights hold the function*.
2. **Gradient-alignment reading.** Per block, $\|\text{PC grad}\|/\|\text{BP grad}\|$ rises $\eta \rightarrow 0.98$ across the homotopy at cosine ≥ 0.998 throughout: the settled local update converges to the full-magnitude BP update in direction and scale. The error field is, to high precision, the adjoint field.
3. **Adjoint sparsity.** Top 5% of cells carry $\approx 90\%$ (KD) / 91.6% (CE) of squared update mass; range 82–97% over fresh samples; depth profile early-to-middle (blocks 2–5: 66% of frontier), dying toward the readout; robust to the KD→CE objective swap; Hill 1.38–2.00, descriptor only.
4. **Deployment.** Deployed artifact = feedforward student exactly (energy minimum at $\varepsilon=0$ absent a task term); post-deployment learning settles under observed-token cross-entropy, the same objective as the alignment probe.
5. **MORK companion.** Full 124M forward in the trie-backed store at 6.06×10^{-6} max relative logit error; KV-cache as incremental view maintenance; 16× blocked-format expression reduction.

2.3 What the revision means for the programme’s three structural changes

The v1.0 changes stand, with sharpened content: absorption returns homotopy-guarded (E1, now with a strong prior that it is clean and a new focus on localizing the R8/R9 cause); emergence remains the keystone (E2, now dual-substrate and startable immediately on adjoint fields); MORK stays pulled forward (E5, with W3 reframed as sparsified-gradient maintenance).

3 Shared platform, notation, and methodology standards

3.1 The rig

Six-layer GPT-2-style transformer on the synthetic compositional grammar, F ground-truth factors ($F=4$ current; **TODO(agent)**: read from rig config). Provided: frozen teacher $\mathcal{T}$; context-signature map $c(x)$; do-operations $x \mapsto x^{\text{do}(j)}$; ID and compositional-shift (CS) splits.

3.2 ePC student parameterisation

Blocks f_ℓ , $\ell = 1, \dots, L$ ($L=6$); $x_\ell = f_\ell(x_{\ell-1}) + \varepsilon_\ell$; $\hat{z} = \text{head}(\text{LN}_f(x_L))$; energy

$$E(\varepsilon) = \sum_{\ell} \frac{1}{2} \|\varepsilon_\ell\|^2 + \mathcal{L}(\hat{z}(\varepsilon)), \quad (1)$$

relaxed T steps from $\varepsilon^{(0)}=0$, weights frozen (assert); local weight objective E_W on settled states; AdamW.

3.3 Fields and substrates (new in v1.1)

Two fields are first-class artifact types, sharing one schema:

- **Settled error field** $\varepsilon(x)$: the terminal-settle error tensors, per block, from an ePC student.
- **Adjoint field** $\delta(x)$: the plain BP adjoints $\partial\mathcal{L}/\partial x_\ell$ of a standard model (teacher or E1 student at $\varepsilon=0$), teacher-free CE, computed by autograd. No homotopy, no settle.

The harness persists both for the shared probe set; every field-level statistic (concentration, depth profile, M1–M5) must run unmodified on either substrate.

3.4 Methodology standards (unchanged from v1.0; mandatory)

Envelope thresholds from $n \geq 3$ replicates, rule fixed in advance; matched seeds and per-seed deltas for all comparisons; pre-registered acceptance criteria in versioned JSON; dual-condition fidelity (clamped and teacher-free) wherever a settle exists in training; per-block reporting for all cell statistics; interventional validation for causal claims (never energy curves); full artifact emission (config hash, seed, commit, JSONL metrics, histograms, standard figures); no silent exclusion — runs launched = runs reported.

4 E1 (complete): record, post-hoc checks, and optional arm E1b

4.1 Status

E1 ran to completion under v1.0 (homotopy re-expression of the six-layer teacher, $n \geq 5$, with the R8/R9 contrast arm). **Agents:** record the branch taken (E1-A/B/C per v1.0 criteria) in the Phase-1 memo; the v1.0 criteria file remains the criteria of record for E1.

4.2 Post-hoc checks (cheap; run on stored artifacts)

P1 (depth profile comparison).

Our top-5% frontier depth profile, terminal settle, KD objective and (if stored; else one re-settle pass) CE objective, against Mesto’s twelve-block profiles rescaled to six blocks. Question: does the early-to-middle locus and readout die-off reproduce at $L=6$ on a synthetic task?

P2 (M0 on E1).

The equivalence diagnostic of §5 applied to the E1 terminal student: per block, cosine and magnitude ratio between the settled error field and the adjoint field of the same student at $\varepsilon=0$ under matched loss, plus top-5% support overlap (Jacc). If M0 is high, our rig confirms the adjoint identification and the E2 adjoint track is the fast path; if M0 is low at six layers, that divergence is itself a finding (the identification is scale- or task-dependent) and both E2 tracks become independently necessary rather than mutually validating.

4.3 E1b (optional, low priority): localize the R8/R9 cause

Mesto’s depth profile makes late-block pile-up look like a property of the training pressure, not of PC. E1b tests this directly: the v1.0 homotopy schedule with the energy-monotonicity term *restored*, matched seeds against both E1 and the contrast arm. Outcome map: if E1b collapses, the energy-monotonicity objective is the cause; if E1b is clean, the cause lies in the schedule/anchor

difference; mixed outcomes localize further by term ablation. Blocks nothing downstream; run when capacity allows.

5 E2 (in flight): dual-substrate frontier stability and fibre emergence

5.1 Question, restated for two substrates

Does field structure — context-stable supports, common invariant subspaces, ground-truth factor alignment — *emerge*, (a) in the settled error field of the E1 student, and (b) in the plain adjoint field of the same/teacher model? Under the adjoint identification the two should agree; M0 measures whether they do, and the adjoint track is runnable on any trained model immediately, no homotopy required.

5.2 New diagnostic M0 (field equivalence)

Per block ℓ , over the probe set: (i) $\cos(\varepsilon_\ell(x), \delta_\ell(x))$ distribution; (ii) magnitude ratio $\|\varepsilon_\ell\|/\|\delta_\ell\|$; (iii) top-5% support overlap $\text{Jacc}(S_\ell^\varepsilon, S_\ell^\delta)$; (iv) agreement of depth profiles. Report against a random-field null. M0 is computed before the rest of the battery is compared across tracks.

5.3 The battery (unchanged definitions, both substrates)

On settled error fields and adjoint fields alike, with matched-size random-cell and shuffled-context nulls throughout:

M1 (support stability).

Context-conditional top- k supports S_c ; within-vs-between split-half contrast $\Delta_{\text{Jacc}} = \mathbb{E}_c[\text{Jacc}(S_c^{(1)}, S_c^{(2)})] - \mathbb{E}_{c \neq c'}[\text{Jacc}(S_c, S_{c'})]$, at $k \in \{1\%, 5\%, 20\%\}$, per block and pooled.

M2 (subspace stability).

Top- r singular subspaces $U_{c,\ell}$ of stacked fields; principal angles, split-half within vs. between contexts, as a function of r .

M3 (JBD recovery).

Context-conditioned second moments $M_{c,\ell} = \mathbb{E}[\phi_\ell \phi_\ell^\top \mid c]$ (with $\phi \in \{\varepsilon, \delta\}$); v0.3 robust JBD per block; recovered blocks, underidentification flags, and the factor-alignment score (MI between block activations and ground-truth factor values, permutation-calibrated).

M4 (interventional fingerprints).

Do-difference fields $\Delta_j(x) = \phi(x^{\text{do}(j)}) - \phi(x)$; selectivity matrix Φ_{jb} (share of $\|\Delta_j\|^2$ in recovered block b); success is a dominant diagonal after Hungarian matching.

M5 (observability floor).

ε_{obs} on the error-field track. Adjoint-track analogue: linear decodability of factor values from δ_ℓ per block, reported as the ceiling against which M1–M4 are read.

Co-location (pre-registered, Prediction 2): freeze, before unblinding M3/M4, the statement that factor-aligned blocks lie in the depth range holding the majority of top-5% mass (per our own P1 profile).

5.4 Branches (per track; the joint read follows M0)

Branch 5.1 (E2-A: emergence). $\Delta_{\text{Jacc}}/\text{M2}$ above nulls, M3 recovers $\geq \lceil F/2 \rceil$ factor-aligned blocks, M4 diagonal-dominant. If on the adjoint track: emergence is a property of trained-transformer credit structure generally, and the recipe simplifies to *identification on adjoint fields*, homotopy optional. If on the error-field track only (M0 low): the settle adds structure the raw adjoint lacks — a genuinely new claim; verify against C2 and re-run with settle-depth ablation before believing it.

Branch 5.2 (E2-B: concentration without structure). Sparsity reproduces, structure sits at null on both tracks. The red regime is real under ideal conditions; the programme’s honest scope contracts to supplied-structure applications; E3’s structured arm uses supplied fibres and measures the cost of imposing structure.

Branch 5.3 (E2-C: blocked by observability). M5 floors near zero. The substrate, not identification, binds; revisit E1 configuration (relaxation depth, loss placement) before concluding on emergence.

6 E3: structured vs. unstructured frontier settle at matched budget

6.1 Question

Does fibre structure buy anything over raw adjoint sparsity — in settle cost, ID quality, or (the real bet) out-of-distribution transfer? v1.1 adds the externally-tuned null (gradient sparsification) and the mechanism ablation (proximal character).

6.2 Conditions (matched total scalar-update budget, metered)

1. **Dense**: full settle (reference).
2. **Top- k** : settle restricted to top- k cells by running magnitude, $k \in \{1\%, 5\%, 20\%\}$.
3. **Sparsified-gradient (new)**: top- k with error feedback / momentum correction in the style of deep gradient compression — the decade-tuned null model for frontier updates. **TODO(agent)**: implement residual accumulation exactly (accumulate unsent mass, not merely re-rank).
4. **Fibre-projected**: updates projected onto E2 fibres (recovered if E2-A, supplied if E2-B), transports across blocks, S3/S4 plasticity gates.
5. **Random- k** (control).
6. **Proximal ablation (new)**: dense settle-based update vs. raw BP step at matched effective step size (calibrate via the E1 magnitude-ratio curve), isolating the implicit regularization of iterative settling. Run as a separate matched-seed pair; report per-step divergence and cumulative trajectory divergence over training.

6.3 Evaluations

V1 (ID quality).

Teacher-free fidelity/task metrics, ID split. Expected: dense $\approx$ top- k $\approx$ sparsified $\approx$ fibre $\gg$ random; ties are the predicted outcome.

V2 (compositional shift).

Same on CS. Discriminating condition: structured routing should degrade more gracefully if fibre structure has content; sparsified-gradient is the bar to beat, not plain top- k .

V3 (transfer of routing).

Routing/gating state frozen on ID, evaluated on CS without re-estimation — does the routing function generalise?

V4 (continual probe).

Sequential factor-restricted streams; off-support drift per condition; C3 (matched-compute BP fine-tune) as the outside comparison. Prediction 1 governs the read: parameterisation alone $\approx$ nothing; protection tracks gates/supports; any residual effect should co-vary with the proximal ablation’s divergence measure.

7 E4 (reframed): test-time information injection vs. the feedforward default

7.1 Reframing

In the corrected picture there is no amortization gap: absent a task term the energy minimizes at $\varepsilon=0$, so **FF is the deployed artifact by construction**. The deployment question is therefore not “does the settle survive removal of the teacher” but purely: *can a settle clamped to test-time information the feedforward pass does not use — observed tokens, symbolic constraints — beat FF, and where?* This isolates the Stage-4 bet exactly: the symbolic cap is the only channel through which inference-time value can enter.

7.2 Conditions (teacher-free evaluation throughout)

1. **FF**: $\varepsilon=0$ feedforward — the deployed default and the baseline to beat.
2. **TC**: teacher-clamped settle — diagnostic upper anchor only (measures headroom available to *any* information injection; never deployable; label it as diagnostic in all outputs).
3. **OT**: observed-token settle (CE on the observed prefix, free continuation).
4. **SC**: symbolic-constraint settle — grammar rules in MeTTa; $\mathcal{L} = \lambda \cdot$ constraint-violation energy queried from the store (batched; in-store path via E5-W4).
5. **OT+SC**: combined.

7.3 Decisive comparison

Per metric and split, the headroom-recovery statistic

$$G = \frac{SC - FF}{TC - FF}, \quad (2)$$

reported as undefined when $TC - FF$ is within noise. The grammar rules are the generative truth of the rig; if $G \approx 0$ here, symbolic clamping will not work anywhere easier, and the deployment story for the whole programme is “PC is a training-time device” — to be stated plainly. If G is material on CS (where FF is weakest), the Stage-4 inversion is vindicated and the symbolic cap

moves up the staging. Secondary: settle steps to criterion (cost of recovered quality); interaction with E3 fibre-projection (do SC corrections localize to M4’s factor blocks?).

8 E5: MORK hosting of the six-layer PC model

Scope unchanged: inference + settle + frontier weight-writes in-store; from-scratch in-store training out of scope.

1. **W1 (forward).** Six-layer forward as MORK rule application, blocked tensor format, dense kernels as sink ops; target $\leq 10^{-5}$ max relative logit error, exact argmax on a decode probe.
2. **W2 (settle in-store).** T -step relaxation as in-store iteration; settled states match reference to pre-registered tolerance.
3. **W3 (frontier writes as IVM), reframed.** Top- k updates as incremental store writes — document and benchmark these as *sparsified-gradient maintenance*: the store-side twin of E3’s sparsified-gradient condition, including the error-feedback accumulator as a store-resident structure. Measure write volume vs. dense; verify reproduction of the reference update to tolerance.
4. **W4 (fibre registration).** E2 fibres (if any) as MeTTa expressions: supports, bases, transports, factor bindings; the substrate E4-SC queries.
5. **W5 (cost accounting).** Wall-clock and expression counts vs. the PyTorch path, reported without spin.

9 Controls and baselines

C1 (BP-residual observability control).

Matched-capacity BP residual/adaptor on the frozen teacher, run on every E3/E4 metric. If C1 matches the PC apparatus everywhere teacher-free, the parameterisation does no deployment work and reports must say so.

C2 (dictionary-learning baseline).

SAE/dictionary learning on *both* substrates (settled error cells and adjoint cells), per block, same probe set; scored on M3 factor alignment and M4 selectivity. If sparse coding matches JBD, the simpler method wins and JBD is repositioned as verification. Mandatory in any E2 write-up.

C3 (matched-compute BP fine-tune).

For all continual claims: matched compute, matched seeds, per-seed deltas.

10 Sequencing (revised) and decision points

Now: E2 continues (both tracks; adjoint track and M0 first, since they are cheap and calibrate the rest); E1 post-hoc checks P1/P2 on stored artifacts; harness deltas of §1.

Next: E3 and E4 in parallel once E2’s M0 and first-pass M1–M4 land (they consume E2’s fibres-or-absence); E5 W1–W2 can start any time (needs only the E1 student); W3 after E3’s sparsified-gradient condition exists (shared implementation); W4 after E2.

Deferred: E1b when capacity allows.

Reporting rule unchanged: each phase closes with an N-series-style memo — established results, boundaries stated with equal force, branch taken and why, predictions checked against their pre-registered statements — before the next phase’s configs freeze.

11 Implementation notes for coding agents

- **Repo.** As v1.0 (`experiments/e1_homotopy/ ...e5_mork/`, `controls/`, `harness/`), plus `harness/fields/` implementing the two-substrate artifact schema (§3) and the M0 evaluator. `TODO(agent)`: reconcile with the tree as built during E1/E2.
- **Criteria files.** Add `criteria_v1.1.json`; do not modify the v1.0 file (E1’s record). E2-in-flight items already run under v1.0 criteria are valid; new items (M0, co-location, adjoint track) run under v1.1.
- **Assertions.** Weights frozen during settle; teacher-free evaluators import no teacher module; E3 budget metering enforced; matched seeds enforced; error-feedback accumulator in the sparsified-gradient condition verified against a dense-update reference on a unit test before campaign runs.
- **Persisted artifacts.** Settled error tensors *and* adjoint tensors for the shared probe set (single source of truth for E2 tracks and C2); per-milestone JSONL; fixed-binning error-mass histograms (Hill and tail sweeps recomputable offline).
- **Standard figures.** v1.0 set, plus: M0 per-block cosine/ratio/ overlap panel; dual-substrate concentration and depth-profile overlays (ours vs. Mesto’s rescaled profiles); the co-location panel (factor-aligned blocks over the frontier depth profile); E3 adds the sparsified-gradient and proximal-ablation curves; E4’s G bars per metric per split with TC labelled diagnostic.
- **Numerical hygiene.** Hill from stored histograms with model-free dispersion errors; never fit a power law; always report the tail-depth sweep next to any fixed-depth α .
- **Failure reporting.** Guard trips and acceptance failures are results: logged, artifacts kept, included in memos; runs launched = runs reported.

A Acceptance criteria (v1.1)

Provisional bars below; final bars frozen per the envelope rule before each campaign, versioned JSON. E1 rows retained for the record (criteria of record: v1.0 file). Deltas per-seed unless noted.

Exp.	Quantity	Provisional bar	Notes
E1	(complete)	per v1.0 file	record branch in memo
E1-P2	M0 support overlap	report; “high” = $\text{Jacc} \geq 0.5$ at top-5%	interpretation gate
E2	M0 cosine	report per block vs. random-field null	compute first
E2	Δ_{Jacc}	$> 3\sigma$ above shuffled-context null	per k , per substrate
E2	factor alignment (M3)	$\geq \lceil F/2 \rceil$ blocks above permutation null	JBD flags clean
E2	fingerprint (M4)	diagonal share $> 2\times$ off-diag mean	after matching
E2	co-location	Prediction 2 checked as stated	frozen pre-unblind
E3	ID tie	top- k , sparsified, fibre within envelope of dense	tie expected
E3	CS separation	fibre vs. <i>sparsified-gradient</i> per-seed delta, sign test	bar raised vs. v1.0
E3	proximal ablation	report per-step and cumulative divergence	mechanism, not bar
E4	G	report; material if > 0.2 on CS	TC diagnostic only
E5	W1 fidelity	$\leq 10^{-5}$ max rel. logit error; argmax exact	Mesto parity
E5	W3 reproduction	sparsified update matches reference to tolerance	incl. accumulator

B Metric definitions gathered

- **Fields.** $\varepsilon_\ell(x)$: settled error tensors. $\delta_\ell(x) = \partial\mathcal{L}/\partial x_\ell$: BP adjoints at $\varepsilon=0$, teacher-free CE, autograd. Shared artifact schema; $\phi \in \{\varepsilon, \delta\}$ below.
- **Squared update mass:** $m_i = \mathbb{E}[\phi_i^2]$ over the probe set; per-sample values stored for split-half statistics.
- **Top- k support S_c :** cells in the top k fraction of m_i on inputs with signature c (majority vote across samples).
- **M0:** per block, distributions of $\cos(\varepsilon_\ell, \delta_\ell)$ and $\|\varepsilon_\ell\|/\|\delta_\ell\|$; $\text{Jacc}(S_\ell^\varepsilon, S_\ell^\delta)$ at top-5%; depth-profile agreement.
- Δ_{Jacc} : within-vs-between context split-half Jaccard contrast (§5).
- **Effective rank (PR):** $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ of per-block settled-state covariance spectra.
- **Hill at depth k :** $\hat{\alpha}^{-1} = \frac{1}{k} \sum_{i=1}^k \ln(m_{(i)}/m_{(k)})$ on order statistics; model-free dispersion error; descriptor, not exponent.
- **Fingerprint selectivity:** $\Phi_{jb} = \mathbb{E}[\|P_b \Delta_j\|^2] / \mathbb{E}[\|\Delta_j\|^2]$, Hungarian-matched before diagonal comparison.
- G : $(\text{SC} - \text{FF}) / (\text{TC} - \text{FF})$ per metric per split; undefined when the denominator is within noise.
- **Proximal divergence:** per-step $\|\Delta W_{\text{settle}} - \Delta W_{\text{BP}}\| / \|\Delta W_{\text{BP}}\|$ at matched effective step size, and cumulative parameter-trajectory distance over a training run.